SIMATS ENGINEERING

ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

Name	Divyasree M
Register Number	192524098

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 50**
Question Count: 5

Code of Conduct

I Divyasree.M/192524098 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: ____________________

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

Definition of Cloud Computing

Cloud computing is the delivery of computing services such as servers, storage, databases, networking, software, analytics, and artificial intelligence over the internet ("the cloud"). Instead of purchasing and maintaining physical hardware, users can access these resources on demand from cloud service providers and pay only for what they use. This provides flexibility, scalability, cost savings, and improved accessibility.

Four Characteristics of Cloud Computing

1. On-Demand Self-Service

Cloud users can provision computing resources such as virtual machines, storage, and databases whenever required without needing manual assistance from the service provider.

Example:
A developer launches a virtual server on Amazon EC2 within minutes using a web console.

2. Broad Network Access

Cloud services are accessible over the internet using standard devices such as laptops, desktops, smartphones, and tablets from any location with an internet connection.

Example:
Students can access files stored in Google Drive from home, college, or a mobile phone.

3. Resource Pooling

Cloud providers use a shared pool of computing resources to serve multiple customers (multi-tenancy). Resources are dynamically assigned based on user demand while keeping customer data isolated and secure.

Example:
Microsoft Azure hosts virtual machines for thousands of organizations on the same physical infrastructure.

4. Rapid Elasticity and Scalability

Cloud resources can be increased or decreased automatically according to workload demands. This allows organizations to handle sudden traffic spikes without purchasing additional hardware.

Example:
An online shopping website automatically increases server capacity during festival sales and reduces it afterward.

Feature	Cloud Computing	Traditional On-Premise Computing
Infrastructure	Managed by cloud provider	Managed by organization
Cost	Pay-as-you-use model	High upfront hardware investment
Resource Provisioning	Instant and automated	Manual installation and configuration
Accessibility	Anywhere through internet	Mostly limited to local network
Scalability	Automatic and flexible	Slow and hardware dependent
Maintenance	Provider handles updates and maintenance	Organization responsible for maintenance

Real-World Examples

- **Amazon Web Services (AWS):** Provides virtual servers, storage, and databases on demand.
- **Microsoft Azure:** Offers scalable cloud infrastructure and AI services.
- **Google Cloud Platform (GCP):** Provides cloud storage, computing, and machine learning services.
- **Google Drive:** Enables users to store and access files from any device connected to the internet.

Conclusion

Cloud computing has transformed the way organizations use IT resources by providing flexible, scalable, and cost-effective services over the internet. Features such as on-demand self-service, broad network access, resource pooling, and rapid elasticity clearly distinguish cloud computing from traditional on-premise computing. These characteristics improve resource utilization, reduce operational costs, and enable businesses to respond quickly to changing demands.

2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

Introduction

Cloud computing did not emerge suddenly; it evolved through several technological advancements over many decades. Improvements in computer hardware, networking, distributed computing, virtualization, and internet software gradually transformed traditional computing into today's scalable and on-demand cloud platforms. Each stage addressed the limitations of the previous generation and contributed to the development of modern cloud services.

Evolution of Cloud Computing

1. Hardware Evolution (1950s–1970s)

Early computers were large, expensive mainframes shared by multiple users through terminals. Organizations had to purchase and maintain their own hardware, making computing costly and accessible only to large institutions.

Key Developments

- Mainframe computers
- Time-sharing systems
- Centralized computing
- Early networking concepts

Contribution to Cloud Computing

- Introduced the idea of sharing computing resources among multiple users.
- Improved utilization of expensive hardware.
- Formed the foundation for resource sharing in cloud environments.

Example: IBM Mainframe Systems.

2. Distributed Computing (1980s–1990s)

Instead of relying on a single computer, computing tasks were distributed across multiple interconnected systems. This increased processing power, fault tolerance, and reliability.

Key Developments

- Client-server architecture
- Distributed systems
- Parallel computing
- Cluster computing

Contribution to Cloud Computing

- Enabled multiple computers to work together.
- Improved scalability and availability.
- Laid the groundwork for distributed cloud infrastructures.

Example: University research clusters and enterprise server farms.

3. Virtualization Technology (Late 1990s–2000s)

Virtualization allowed multiple virtual machines (VMs) to run on a single physical server, with each VM operating independently.

Key Developments

- Hypervisors
- Virtual Machines
- Server consolidation
- Efficient resource utilization

Contribution to Cloud Computing

- Improved hardware utilization.
- Reduced infrastructure costs.
- Enabled rapid provisioning and isolation of computing environments.
- Made Infrastructure as a Service (IaaS) possible.

Example: VMware, Microsoft Hyper-V.

4. Internet Software Evolution (2000s–Present)

The widespread availability of the internet enabled software and services to be delivered online instead of being installed on individual computers.

Key Developments

- Web applications
- Web services (SOAP and REST)
- APIs

- Service-Oriented Architecture (SOA)
- Broadband internet

Contribution to Cloud Computing

- Allowed applications to be accessed from anywhere.
- Enabled communication between different systems through APIs.
- Supported Software as a Service (SaaS), Platform as a Service (PaaS), and cloud-based collaboration.

Examples:

- Gmail
- Google Docs
- Microsoft 365
- Salesforce

5. Modern Cloud Platforms (Present)

Combining virtualization, distributed computing, high-speed internet, and web technologies resulted in highly scalable cloud platforms.

Characteristics

- On-demand self-service
- Resource pooling
- Rapid elasticity
- Broad network access
- Pay-as-you-go pricing
- Global availability
- Automatic scaling

Examples

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform (GCP)

How These Changes Enabled Modern Cloud Platforms

Evolution Stage	**Major Advancement**	**Contribution to Modern Cloud Platforms**
Hardware Evolution	Mainframes and Time Sharing	Introduced shared computing resources
Distributed Computing	Multiple interconnected computers	Improved scalability and reliability
Virtualization	Virtual Machines and Hypervisors	Enabled efficient resource allocation and Infrastructure as a Service (IaaS)
Internet Software Evolution	Web applications, APIs, REST, SOAP	Enabled remote access, interoperability, and Software as a Service (SaaS)
Modern Cloud Platforms	Integration of all technologies	Provides scalable, secure, on-demand cloud services worldwide

Real-World Example

Consider an e-commerce company:

- **Hardware Evolution:** Initially used one physical server to host the website.
- **Distributed Computing:** Added multiple servers to manage increasing customer traffic.
- **Virtualization:** Consolidated servers into virtual machines to reduce hardware costs.
- **Internet Software Evolution:** Adopted REST APIs and web services for payments, inventory, and shipping.
- **Modern Cloud Platform:** Migrated to AWS or Azure, where servers automatically scale during peak shopping seasons and scale down afterward, reducing costs while maintaining performance.

Conclusion

Cloud computing evolved through continuous advancements in hardware, distributed computing, virtualization, and internet software technologies. Hardware evolution introduced shared computing, distributed systems improved scalability, virtualization maximized resource utilization, and internet software enabled web-based service delivery. Together, these innovations created today's cloud platforms, which offer flexible, scalable, reliable, and cost-effective computing services to users worldwide.

3. Explain the Role of Server Virtualization in Cloud Computing. Differentiate Between Virtualization as an Enabling Technology and Cloud Computing as a Service Model.

Introduction

Server virtualization is one of the core technologies that made cloud computing possible. It allows multiple virtual servers, known as **Virtual Machines (VMs)**, to run independently on a single physical server. Each VM has its own operating system, applications, and resources, while sharing the same hardware through a software layer called a **hypervisor**.

Role of Server Virtualization in Cloud Computing

1. Efficient Resource Utilization

Virtualization enables multiple virtual machines to share a single physical server, ensuring better utilization of CPU, memory, and storage.

Example: One physical server can simultaneously host a Linux VM, a Windows VM, and a database server.

2. Rapid Resource Provisioning

New virtual machines can be created within minutes, allowing cloud providers to deliver computing resources quickly.

Example: AWS EC2 allows users to launch virtual servers on demand.

3. Scalability

Virtual machines can be added or removed according to workload, supporting automatic scaling in cloud environments.

Example: During online shopping festivals, cloud providers automatically deploy additional virtual servers to handle increased traffic.

4. Isolation and Security

Each virtual machine operates independently. If one VM fails or is attacked, other VMs remain unaffected.

Example: A software testing VM crashing does not impact the production VM on the same server.

Virtualization vs Cloud Computing

Feature	**Virtualization**	**Cloud Computing**
Definition	Technology that creates virtual versions of physical resources	Service model that delivers computing resources over the internet
Purpose	Improve hardware utilization	Deliver IT services on demand
Dependency	Foundation technology	Built using virtualization and other technologies
Internet Required	Not necessary	Usually internet-based
Resource Ownership	Organization owns infrastructure	Cloud provider manages infrastructure
Example	VMware, Hyper-V	AWS, Microsoft Azure, Google Cloud

Real-World Example

A company installs **VMware** to create ten virtual servers on one physical machine (Virtualization). Later, it migrates its applications to **Microsoft Azure**, where virtual servers are provided as an online service (Cloud Computing).

Conclusion

Server virtualization enables efficient use of hardware resources by creating multiple virtual machines on a single physical server. While **virtualization is the enabling technology**, **cloud computing is the service model** that uses virtualization, networking, automation, and internet technologies to provide scalable, on-demand computing services.

4. Compare SOAP-Based and REST-Based Web Services in Terms of Communication Style, Data Exchange, and Suitability for Cloud Applications.

Introduction

Web services allow different applications to communicate over a network. The two most widely used approaches are SOAP (Simple Object Access Protocol) and REST (Representational State Transfer). Both enable system integration but differ significantly in communication methods, message formats, and performance.

SOAP-Based Web Services

SOAP is a protocol that uses XML messages and follows strict communication standards. It provides high security, reliability, and transactional support.

Characteristics

- Protocol-based
- XML data format only
- Highly secure
- Supports ACID transactions
- Platform independent

Example: Banking systems and online payment gateways.

REST-Based Web Services

REST is an architectural style that uses standard HTTP methods such as GET, POST, PUT, and DELETE. It supports multiple data formats, especially JSON.

Characteristics

- Lightweight
- Uses HTTP protocol
- Supports JSON, XML, HTML, and plain text
- Faster performance
- Easy integration

Example: Google Maps API, Twitter API, GitHub API.

Comparison Between SOAP and REST

Feature	SOAP	REST
Type	Protocol	Architectural Style
Communication Style	Strict messaging protocol	HTTP request-response
Data Exchange	XML only	JSON, XML, HTML, Text
Performance	Slower due to XML overhead	Faster and lightweight
Security	WS-Security supported	HTTPS, OAuth, JWT
Flexibility	Less flexible	Highly flexible
Ease of Development	Complex	Simple
Cloud Suitability	Enterprise systems requiring high security	Modern cloud-native applications and APIs

Real-World Examples

- **SOAP:** Online banking transactions requiring strong security and reliability.
- **REST:** Mobile applications accessing weather information using REST APIs.

Conclusion

SOAP is ideal for enterprise applications that require strong security, reliability, and transactional integrity. REST is preferred for modern cloud applications because it is lightweight, faster, scalable, and easier to integrate with web and mobile applications.

5. Differentiate IaaS, PaaS, SaaS, and XaaS with One Suitable Real-World Example for Each Model.

Introduction

Cloud service models define how computing resources are delivered to users. The major models are **Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Software as a Service (SaaS), and Anything as a Service (XaaS).** Each model offers a different level of management responsibility between the cloud provider and the user.

1. Infrastructure as a Service (IaaS)

IaaS provides virtualized computing infrastructure such as servers, storage, networking, and operating systems over the internet.

Features

- Virtual machines
- Storage
- Networking
- User manages applications and operating system

Example: Amazon EC2

2. Platform as a Service (PaaS)

PaaS provides a complete platform for application development, testing, and deployment without managing the underlying infrastructure.

Features

- Development tools
- Runtime environment
- Databases
- Middleware

Example: Google App Engine

3. Software as a Service (SaaS)

SaaS provides fully functional software applications over the internet. Users simply access the software using a web browser.

Features

- No installation required
- Accessible anywhere
- Automatic updates

Example: Microsoft 365

4. Anything as a Service (XaaS)

XaaS is a broad category that includes all cloud-based services such as Storage as a Service, Database as a Service, AI as a Service, Security as a Service, and many more.

Features

- Flexible service delivery
- Subscription model
- Covers all cloud services

Example: OpenAI API (AI as a Service)

Comparison of Cloud Service Models

Service Model	What It Provides	User Manages	Cloud Provider Manages	Real-World Example
IaaS	Virtual servers, storage, networking	OS, applications, data	Hardware, virtualization	Amazon EC2
PaaS	Development platform	Applications and data	Infrastructure, OS, runtime	Google App Engine
SaaS	Complete software application	Only application usage	Everything else	Microsoft 365
XaaS	Any cloud-based service	Depends on service	Most infrastructure and service management	OpenAI API (AIaaS)

Real-World Scenario

A software company develops an online shopping application:

- Uses Amazon EC2 (IaaS) to rent virtual servers.
- Uses Google App Engine (PaaS) to develop and deploy the application.
- Uses Microsoft 365 (SaaS) for employee email and collaboration.
- Uses the OpenAI API (AI as a Service under XaaS) to power a customer-support chatbot.

Conclusion

IaaS, PaaS, SaaS, and XaaS provide different levels of cloud services based on user requirements. IaaS offers infrastructure, PaaS offers a development environment, SaaS delivers ready-to-use software, and XaaS extends the cloud model to virtually any IT service. Choosing the right model depends on the organization's technical needs, budget, and level of control required.